Toward Opportunistic Compression and Transmission for Private Car Trajectory Data Collection

Jie Chen, Zhu Xiao, *Member, IEEE*, Dong Wang, *Member, IEEE*, Daiwu Chen, Vincent Havyarimana, Jing Bai, *Member, IEEE*, and Hongyang Chen, *Senior Member, IEEE*

***Abstract*— The advances in vehicle location service and communication techniques have generated massive spatial-temporal trajectory data, which has caused the crises of storage and communication in the vehicle trajectory data center. In this paper, we propose a novel opportunistic compression and transmission-based long short-term memory method, namely, OCT-LSTM, with aims of reducing trajectory transmission overhead and storage cost. We first present a low-cost vehicle location device for trajectory real-time collection of private cars. Within the proposed OCT-LSTM, we introduced a map-matching method based on MIV-matching which reduces sampling errors of raw trajectories. Then, we present a spatial-temporal transformation method to divide the trajectory data into two parts, i.e., spatial path and time-distance parts, and realize the compression operation separately. Similar movement patterns are repeated and randomly present in trajectories of private cars. In this paper, we train the LSTM model to remember and predict these repetitive movement patterns through historical trajectories. An opportunistic transmission of trajectory data from the vehicle terminal to the data center was designed, which can dramatically decrease the transmission overhead. The proposed OCT-LSTM not only realizes real-time trajectory preprocessing and compressing but also ensures high trajectory compression ratio. To validate the performance of the OCT-LSTM, we collect a large-scale private car trajectory data from real urban environments. The experiments verify the compression ratio effectiveness and time-delay superiority of the proposed methods.**

***Index Terms*— Trajectory compression, trajectory real-time transmission, long short-term memory, private car.**

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J. Chen and Z. Xiao are with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China, and also with the State Key Laboratory of Geo-Information Engineering, Xi'an 710068, China (e-mail: chemjay@hnu.edu.cn; zhxiao@hnu.edu.cn).

D. Wang is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha 410082, China (e-mail: wangd@hnu.edu.cn).

D. Chen is with the College of Information, Hunan University of Humanities, Science and Technology, Loudi 417000, China (e-mail: daiwuchen@huhst.edu.cn).

V. Havyarimana is with the Department of Applied Sciences, Ecole Normale Supérieure de Bujumbura, Bujumbura 6983, Burundi (e-mail: havincent12@gmail.com).

J. Bai is with the Key Laboratory of Intelligent Perception and Image Understanding, School of Artificial Intelligence, Ministry of Education, Xidian University, Xi'an 710071, China (e-mail: baijing@mail.xidian.edu.cn).

H. Chen is with the Institute of Industrial Science, The University of Tokyo, Tokyo 153-8505, Japan (e-mail: dr.h.chen@ieee.org).

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I. Introduction

THE explosive growth of vehicle has caused many problems in our urban life, such as the traffic jam, road accidents and exhaust gas pollution [1]–[4]. Nowadays, the trajectory data collection offers us unprecedented information to understand travel behavior, fostering a broad range of applications in location-based social networks, intelligent transportation systems, and urban computing [5]–[8].

However, the emergence of large amounts of trajectories has caused the crises of storage and communication, especially for the high storage cost and heavy transmission overhead of the data center. For instance, trajectory data collected from 59,700 taxis in Beijing within only 3 months has the size around 1TB and generated almost 100 million times upload [9]. Such a large volume of raw trajectories is limited to provide high resolution trajectory information for better management and services. The real-time requirement in vehicle network is an important factor to ensure the service quality of road vehicles [10]–[12]. In the meantime, many vehicle location services require real-time trajectory collection, such as vehicle surveillance [13], asset tracking [14], fleet management [15]. For example, a vehicle tracking system collects location information of vehicle via GPS (Global Position System) technology and transmit the data in near-real-time via cellular or satellite networks to a computer or data center for evaluation. These applications use a broad concept of real-time, which refers to the provision of a service response within seconds.

In addition, private cars make up the majority of automobiles that drive in the urban road networks [16]. For instance, nationwide ownership of small passenger vehicles had exceeded 60 million by the end of 2016 in China, 92.60% of which were registered by individuals, i.e., private cars. Moreover, in most large cities in China, the private cars make up a continuous growing portion of the automobiles, for example, increased 22.08 million (grown by 15.08%) in 2016. On one hand, trajectories data of private cars greatly reflect

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the people's movement pattern and travel behavior [5], on the other side, the large amount of private cars trajectory data pose an unaffordable burden for data storage and transmission. One of the main goals of this paper is to reduce the transmission overhead for trajectory collection.

In order to reduce the data transmission overhead and storage cost of a large amount of trajectory collection. This paper advances vehicle trajectory data real-time service with the following contributions:

a) We present a low-cost vehicle location device and OCT (Opportunistic Compression and Transmission) strategy for vehicle trajectory collection of private cars. OCT strategy is proposed for the first time, which can compress the vehicle trajectory data and reduce the overhead of data to upload on the data center.

b) We construct LSTM (Long Short-Term Memory) model to train and predict the trajectory of private cars, which can excavate the similar movement pattern from the historical trajectory and predict the future move state of the vehicle. Meanwhile, the quality of the prediction model greatly affects the effect of opportunistic compression, because in this strategy, an accurate prediction means that one time transmission can be reduced. So we want to build the LSTM-based prediction model as accurate as possible.

c) We evaluate the effectiveness and efficiency of our method with extensive experiments and utilize real-life datasets to verify the compression ratio effectiveness and time-delay superiority of the proposed methods. Our method outperforms baselines significantly.

The remainder of this paper is organized as follows. In Section II, the related work is summarized. The preliminary definitions are presented in Section III. In Section IV trajectory collection equipment of private car is described, followed by Section V in which the opportunistic compression method is presented. Numerical results are presented and discussed in Section VI before we conclude the paper in Section VII.

II. Related Work

The rapid increase of GPS devices in recent years is closely integrated into our modern life. One challenge that location-based applications need to overcome is the amount of data that continuous location tracking can yield. Efficient storage and transmission of these data are critical to their success, especially for vehicle location service which has restricted transmission cost, time delay, and storage [17]–[19]. In this section, we review two areas related to our study: trajectory compression methods, and LSTM (Long Short Term Memory) methods for trajectory prediction.

Several trajectory compression methods have been proposed to simplify GPS trajectory. These methods these methods can be grouped into two categories: off-line compression and online compression. One of the most classical trajectory compression algorithms called Douglas-Peucker algorithm was presented in 1973 [20], which recursively selects the point whose perpendicular distance is greater than given threshold until all points reserved to meet the condition. The disadvantage of the Douglas-Peucker algorithm is that it runs off-line and requires the whole trajectory. A generic sliding window algorithm [21] is to append the points to a growing sliding window and continue to grow the sliding window until the approximation error exceeds a predefined error tolerance. The sliding-window algorithm can run online, but its worst-case runtime is still $O(nL)$ where L is the maximum sliding window size. The Spatial QUalIty Simplification Heuristic (SQUISH) method based on the priority queue data structure proposed by Muckell *et al.* [22] prioritizes the most important points in a trajectory. It achieves relatively low runtime, high compression ratios, and small trajectory errors. However, its disadvantage was that it could not guarantee trajectory errors to be within a specific bound. An extended version called SQUISH-E (Spatial QUalIty Simplification Heuristic-Extended) which has the flexibility of tuning compression with respect to compression ratio and error [23]. While the compression ratio-bound flavor of SQUISH-E can run online, the error-bound version runs off-line only. Multiple examples of trajectory compression in road networks exist in the literature [24]–[27], which first projects the trajectory onto the road network via existing map-matching algorithms and then invokes compression algorithms to compress the trajectory. These algorithms, however, do not apply to our scenario as they only run off-line. PRESS (Paralleled Road-Network-Based Trajectory Compression) [28], a classical map-matching based trajectory compression method, adopts the shortest path compression and develops an encoding algorithm to further compress the trajectory. It also modifies an error-bounded algorithm for temporal compression. Although PRESS focuses on batch model based trajectory compression, the compression procedure scans the spatial path and temporal sequence from head to tail without tracing back. This means PRESS can be adapted to online compression. In addition, a model-based trajectory compression method is also used for trajectory compression. Hung and Peng proposed a framework, called as Model-based Data Collection (MDC) [29], which executed at the server and vehicle side collaboratively. In the vehicle side, given a series of GPS data points, model functions are derived to represent the raw GPS data points. Each vehicle reports some coefficients that describe its movements instead of reporting all position information. The MDC framework has a similar concept of our paper, which utilizes the models both on the vehicle side and the server side to reduce the transmission of trajectory data. Although MDC does not have the compression processing such as map matching, stop points discarding, and shape filling, etc. The main difference is that MDC does not support real-time trajectory compression. This is because the MDC replaces the trajectory transmission by the parameters fitted by the synchronization model, while OCT-LSTM uses the prediction model to reduce the synchronization, which can update the model information for each sampling interval.

The Long Short-Term Memory (LSTM) network was developed to overcome the shortcoming of RNN, i.e., vanishing and exploding gradient problem [30]. LSTM networks have been shown to successfully learn and generalize the properties of isolated sequences like human trajectory [31] and surround vehicles [32]. Khosroshahi *et al.* [32] proposed LSTM

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for detecting the surrounding vehicles' trajectory types. This inspired us to use LSTM to predict the future trajectory of vehicles to reduce the transmission of trajectory data. Wong *et al.* [33] proposed LSTM-PPM (Long Short Term Memory with Periodic Pattern Mining) utilizes deep learning and periodic pattern mining for long-term prediction of user locations. However, this method is concerned with the prediction of the user's staying points, not the prediction of the vehicle movement trajectory. In summary, these methods compress in the form of simplifying the number of trajectory points. Although the form of online compression is supported, a small portion of the track (sliding window) still needs to be processed to update the trajectory information, which restricts the timeliness of the trajectory update.

Except for the encoding based compression methods, most of the current trajectory compression methods are based on the strategy of simplifying the number of trajectory points. Even for online compression methods, a small portion of the trajectory (sliding window) still needs to be processed to update the trajectory information, which restricts the timeliness of the trajectory update. In this paper, we propose an OCT strategy that uses two predictive models on the vehicle side and the server side to reduce the transmission of trajectory data. The transmission process only occurs when the difference between the predicted value and the real value of trajectory is greater than error-bound. This OCT strategy is different from the previous trajectory compression methods and can realize nearly real-time compression.

III. Definitions

To describe the concept of Opportunistic Compression and Transmission (OCT) conveniently, some important definitions of this paper are defined as follow.

Def. 1 (Road Network): The topology of real-world roads are defined as a directed graph $G = (V; E)$, where $V = \{v_1, v_2, , \ldots, v_l\}$ is the set of intersections and $E = \{e_1, e_2, \ldots, e_m\}$ is the set of road segments, a road segment $e_i = \{c_1, c_2, \ldots, c_k\}$ contains several coordinated point to represent the geometric shape of a road segment. The $c_k = (lon_k, lat_k)$ is the position in 2D Euclidean space, where lon_k and lat_k represent longitude and latitude, respectively. As shown in Fig. 1 (a), The gray line is a road network composed of V and E.

Def. 2 (Raw Trajectory): A raw trajectory is a sequence of n triples in the form of $T_{raw} = \{p_1, p_2, \ldots, p_n\}$, where $p_n = (lon_n, lat_n, t_n)$, (lon_n, lat_n) is the position in two-dimensional Euclidean space and t_n is the sampling moment. As shown in Fig. 1(a), the black dotted line points is T_{raw}.

Def. 3 (Matched Trajectory): A matched trajectory is $T_{matched} = \{m_1, m_2, \ldots, m_n\}$, which is the raw trajectory T_{raw} be aligned to the existing road segments E in the road network G by map-matching algorithms, where $m_n = (lon'_n, lat'_n, t_n, id)$. t_n represent the sampling moment. lon'_n, lat'_n represent the matched point's longitude and latitude, respectively. id represent the mapping road segment ID.

Def. 4 (Path Sequence): Given a Path Sequence $Path = \{id_1, id_2, \ldots, id_n\}$ its represent the road segment ID has been passed, where id_n represents the corresponding road segment ID.

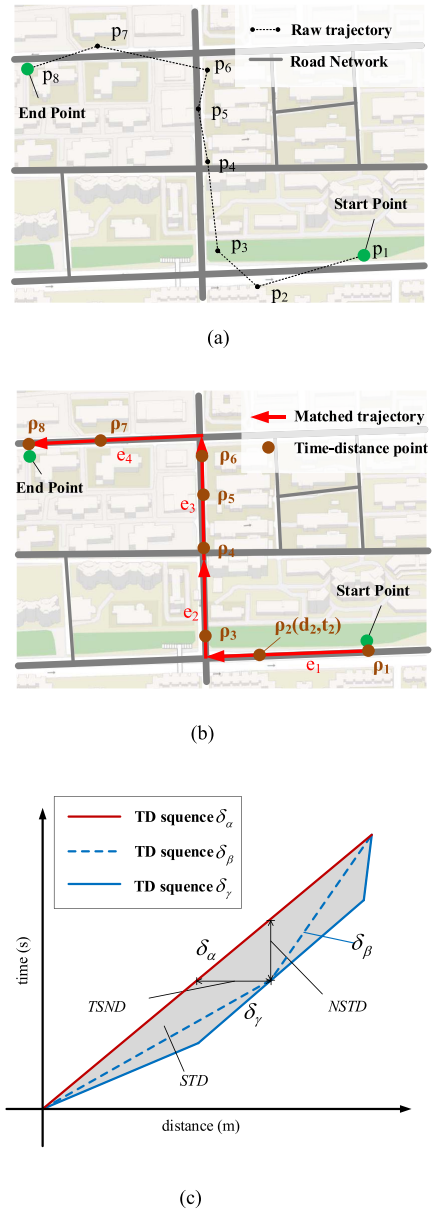

Fig. 1. The illustration of definitions. (a) Raw trajectory and Road Network. (b) Matched trajectory and Time-distance point. (c) The Spatial-Temporal Difference of TD sequences.

Def. 5 (Time-Distance Sequence): A Time-Distance (TD) sequence $\delta = \{\rho_1, \rho_2, \ldots, \rho_n\}$ represents the temporal information of $T_{matched}$, after trajectory spatial-temporal transformation. As shown in Fig. 1(b), where $\rho_n = (d, t)$ is time-distance point, d is the distance from the trajectory start point and t is the sampling moment at d.

Def. 6 (Spatial-Temporal Difference): A Spatial-Temporal Difference (STD) $\psi(\delta_\alpha, \delta_\beta)$ represents the spatial-temporal of two time-distance sequence δ_α and δ_β, its physical meaning is the internal area between two time-distance curves (described by time-distance sequence) in the Time & Distance plane, as shown in Fig. 1(c). The STD is an improved version of Network Synchronous Time Difference (NSTD) and Time Synchronous Network Distance (TSND) metrics in [28].

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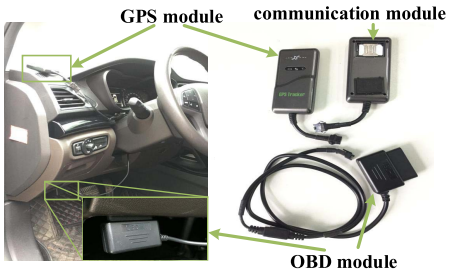

Fig. 2. The picture of vehicle location terminal.

IV. The Trajectory Collection Equipment

In this paper, we present a low-cost integration device to implement private cars trajectory collection. As shown in Fig. 2, the vehicle location terminal mainly includes three modules, namely GPS module, OBD (On-Board Diagnostics) module and the communication module. The GPS module built-in computing, small storage and GPS chips which can realize calculation, storage and localization functions. The OBD module which can read vehicle sensor information from the vehicular OBD interface and supply the power to the terminal. By making use of communication module and LTE (Long Term Evolution) technology, the collected trajectory information can be sent back to our data center real-time. In this terminal, the trajectory data upload frequency can be set in the range from 5 to 1/60 Hz according to the need of service. Meanwhile, the higher upload frequency, the more movement details can be obtained. Inevitably, this leads to heavy transmission overhead and storage cost. To solve this, we propose a novel Opportunistic Compression and Transmission (OCT) strategy for private cars trajectory data, the main idea of which is to reduce both the size of the data transmission and the frequency of uploading by using the trajectory prediction method.

V. Methodology

In this section, we first introduce the overview of OCT-LSTM approach in details. Then, we introduce the preprocessing of the trajectory data in map-matching and spatial-temporal transformation stage. Next, an LSTM predicting model and OCT strategy is introduced, respectively. Finally, an STD algorithm is introduced to measure the temporal and spatial differences of different trajectories.

A. The OCT-LSTM Approach

Fig. 3 provides an overview of the OCT-LSTM approach, which consists of two components: the location terminal in the private car and the data center. First, we obtain the position of the private car through the GPS module, then we align raw trajectories T_{raw} with the road network G by map-matching method [34]. Second, matched trajectory $T_{matched}$ was divided to $Path$ and TD sequence δ though spatial-temporal transformation method. However, time-distance sequence δ still cause high transmission cost, we use LSTM prediction model learned from historical trajectories and OCT strategy to realize opportunistic compression. This LSTM prediction model runs

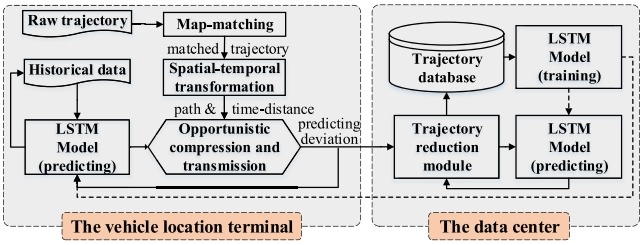

Fig. 3. The overview of OCT-LSTM.

synchronously on both the terminal and the data center side, which can predict the trajectory of the next moment of the vehicle. The opportunistic transmission only happens when there is a large deviation between the predicted value and the real value. The LSTM prediction model is pre-trained at the data center, and then the trained model parameters are synchronized to the vehicle location terminal. As shown in Fig. 3, the black dotted arrow indicates that the parameters of LSTM model are transmitted back to the vehicle through the network's downlink bandwidth. Meanwhile, a maximum spatial-temporal difference uses to evaluate the predicting deviation. In this case, the opportunistic compression strategy realizes the reduction of transmission overhead and storage cost.

B. The Map-Matching Stage

We utilize a map-matching method in order to cope with the sparsity and noise is a serious problem in the raw trajectory data. The map-matching is an effective way to solve these problems. Our goal in the map-matching stage is to estimate a $T_{matched}$ from a raw GPS trajectory T_{raw} which is as similar as possible to its actual path, as shown in Fig. 1. We then map the T_{raw} to $T_{matched}$ by a map-matching approach called MIV-matching (Min-miscalculation Interactive-Voting based Map-matching) which previously proposed in [34]. Vehicles generally travel on the road, and through the MIV-matching, a part of the GPS positioning error can be corrected. It should be pointed out that the error rate of the MIV-matching method is very low under the high frequency GPS sampling, hence the accuracy and error correction of the matching method is not discussed in this paper.

C. The Spatial-Temporal Transformation Stage

In the spatial-temporal transformation stage, we first divide T_{raw} into TD sequence δ and path sequence $Path$, which is a dimensionality reduction process essentially. The path sequence $Path$ is easy to be compressed in a spatial dimension. The key idea in spatial compression is the dictionary compression technique, which use road segment ID sequence to represent the path which segment has been passed. By using the dictionary compression technique, we can use only one id to record the trajectory passing this road segment, where $e = \{c_1, c_2, \ldots, c_k\}$. Therefore, this can effectively compress the spatial information of the trajectory.

In addition to the division of the space-time dimension, two steps called Stop Points (SPs) Discarding and Shape

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Algorithm 1: Stop Points Discarding & Shape Filling

Input: $\delta_p = (\rho_{i-n}, \ldots, \rho_{i-1})$, ρ_i, id, G
Output: $\widehat{\delta}$

1: Initialization: $\widehat{\delta} \leftarrow \delta_p$; $flag \leftarrow True$
2: /*Stop Points Discarding*/
3: **if** isMoving(δ_p, ρ_i) **then**
4: $\quad flag \leftarrow False$
5: **end if**
6: /*Shape Filling*/
7: **if** $flag == True$ **then**
8: $\quad$ **while** isMissingShape($\widehat{\delta}$, ρ_i, id, G) **do**
9: $\quad\quad d_{missing} \leftarrow$ Get d of missing road coordinates
10: $\quad\quad t_{missing} \leftarrow$ Get t by linear interpolation
11: $\quad\quad \rho_{missing} \leftarrow (d_{missing}, t_{missing})$
12: $\quad\quad$ Insert $\rho_{missing}$ into $\widehat{\delta}$
13: $\quad$ **end while**
14: $\quad$ Insert ρ_i into $\widehat{\delta}$
15: **end if**
16: **return** $\widehat{\delta}$

Filling, are also carried out in the spatial-temporal transformation stage, as shown in Algorithm 1. We use $\delta_p = (\rho_{i-n}, \ldots, \rho_{i-1})$, ρ_i, id, E as input data to detect SPs and the missing shape of trajectory, where δ_p is $n-1$ previous time-distance sequence has completed SPs Discarding and Shape Filling. ρ_i is time-distance point of the current sample, id is the segment id of the current position, and G is the road network. In the SPs discarding step, several Stop Points have been discarded. In this step, if point ρ_i is judged to be a stop point, set the $flag$ to $False$, as shown in lines 3 to 5. Raw trajectory T_{raw} is a discrete sampling of moving locations of vehicle. This kind of discrete recording will inevitably result in the loss of information between the adjacent sampling intervals. Since the vehicle is traveling on the road network in general, some missing points can be filled according to the road network in the Shape Filling step, as shown in lines 7 to 15. However, the representation of time-distance sequence δ is still storage consuming, as it suffers from the same scale as T_{raw}. It's need further compressed by OCT-LSTM strategy.

D. The LSTM Predicting Model

When the vehicle passes through the same segment of the road, because of the road network structure, there will be repetitive movement patterns, such as waiting for traffic lights, right turn, start acceleration and so on. Fig. 4 is an example of the conversion to TD sequence form via the same path historical trajectory. For better observation and comparison, the starting time of these TD sequences is reduced to zero. The black dotted frame shows the vehicle waiting and decelerating as it turns left at a traffic light junction. Very similar movement patterns appear on the historical TD sequences in the same road segment. In this paper, we use the LSTM model to memorize and identify this repetitive movement pattern. The LSTM consists of the cell memory that stores the summary of the past input TD sequence, and the gating mechanism by

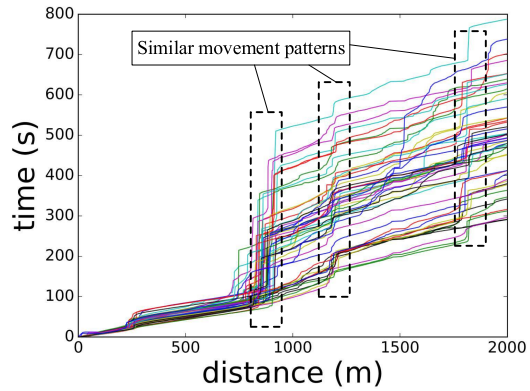

Fig. 4. The illustration of TD sequence data.

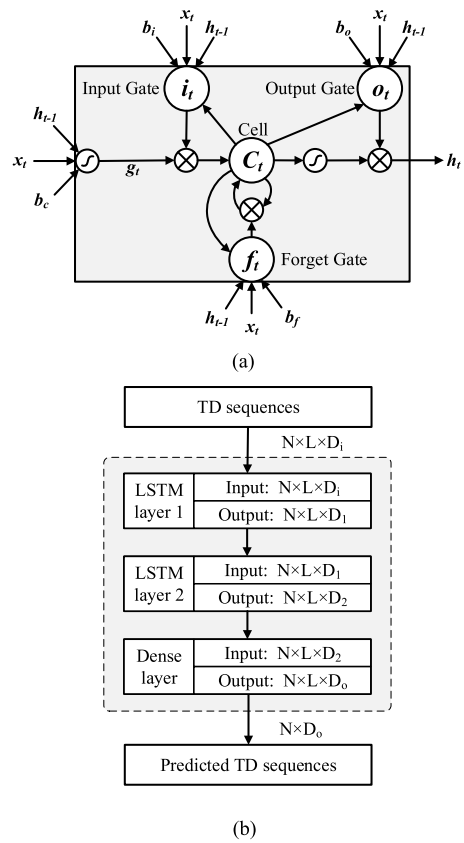

Fig. 5. The LSTM cell and model structure. (a) The illustration of LSTM cell. (b) The illustration of LSTM model.

which the information flow between the input, output, and cell memory are controlled, as shown in Fig. 5(a). Let C_t be the state of the memory cell at the current time step t. Then, C_t is updated by the following recursive equations

$$i_t = \sigma\left(W_{xi}x_t + W_{hi}h_{t-1} + b_i\right), \tag{1}$$
$$f_t = \sigma\left(W_{xf}x_t + W_{hf}h_{t-1} + b_f\right), \tag{2}$$
$$o_t = \sigma\left(W_{xo}x_t + W_{ho}h_{t-1} + b_o\right), \tag{3}$$
$$g_t = \tanh\left(W_{xc}x_t + W_{hc}h_{t-1} + b_c\right), \tag{4}$$
$$C_t = f_t \odot C_{t-1} + i_t \odot g_t, \tag{5}$$
$$h_t = o_t \odot \tanh\left(C_t\right), \tag{6}$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is sigmoid function, $x \odot y$ is element-wise product, b_i, b_f, b_o, b_c are bias vectors, i_t is input gating

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vector, i_t is input gating vector, f_t is forget gating vector, o_t is output gating vector, g_t is state update vector, h_t is output hidden state vector, and W_{xi}, W_{hi}, W_{xf}, W_{hf}, W_{xo}, W_{ho}, W_{xc}, W_{hc} are weight matrixes for linear transformation.

In this paper, the LSTM spatial-temporal prediction model is trained by historical trajectory data, which has been converted to the corresponding TD sequence form. Then we implement the same LSTM prediction model on both the data center side and the terminal, which will automatically predict the trajectory values at each moments. To better learn the temporal representations, two layers of LSTM and one dense layer were stacked on top of each other. The last layer outputs a vector with the predicted values, as shown in Fig. 5(b). Where the parameter N is the number of samples, L is the length of timesteps, D_i is input dimension, D_o is output dimension, D_1 and D_2 is the LSTM cell number of each LSTM hidden layer. In our experiment, we set $L = 20$, representing 20 TD points at the preceding moment. Set $D_i = D_o = 1$ and $D_1 = D_2 = 50$. These parameters can be adjusted according to actual requirements. Due to the recurrent nature of LSTM, even a single layer of LSTM nodes can be considered as a "deep" neural network. Although such layers may theoretically be stacked in a fashion similar to convolutional neural networks to learn higher-level features. However, considering the efficiency of the vehicle positioning, we adopt a two-layer structure in the paper.

E. The OCT Strategy

In OCT strategy, the main idea is to reduce the number of data uploads by using the LSTM prediction models both on the vehicle side and the server side. There two situations in this OCT strategy.

(1) The first situation is the actual trajectory (in the $Path$ and TD sequence δ form) of the vehicle is consistent with the predicted value (or the prediction deviation is smaller than the predetermined error-bound). In this situation, we do not carry out communication transmission, and the data center just uses the predicted value obtained by the prediction model.

(2) The second situation is the prediction deviation is larger than the predetermined error-bound, the terminal calculates the predicting deviation, which between the prediction value and the real value, and then upload it. At this point, the data center receives the deviation and corrects the prediction model. Note that the initial location and time of the vehicle is delivered directly to the data center.

In the OCT-LSTM method, we use the historical trajectory data to train the LSTM model and expect to extract the moving information of the vehicles. However, the LSTM model cannot be trained when we encounter a section without historical data. At this time, we use the linear prediction model of the average velocity of the previous n sample points, in order to distinguish from the OCT-LSTM, we simply call it OCT. As private cars often travel regularly, according to the PriTra dataset of this paper, about 71.9% of the prediction model can be applied to the OCT-LSTM pattern.

F. The Spatial-Temporal Difference & Encode Methods

1) Spatial-Temporal Difference: At present, there are many methods to measure the trajectory similarity. Network Synchronous Time Difference (NSTD) and Time Synchronous Network Distance (TSND) are metrics to measure the temporal and spatial differences of time-distance sequences [28]. However, NSTD and TSND still exist the situation that cannot measure δ difference effectively, as shown in Fig. 1(c), the NSTD and TSND of δ_α and δ_β are the same as δ_α and δ_γ. In order to better distinguish the spatial-temporal difference between two time-distance sequence δ_α and δ_β, we proposed STD (Spatial-Temporal Difference) $\psi(\delta_\alpha, \delta_\beta)$ for measurement, which is the internal area between δ_α and δ_β in the Time & Distance plane. The pseudocode of calculate Spatial-Temporal Difference is shown in Algorithm 2.

Algorithm 2: Calculate the STD $\psi(\delta_\alpha, \delta_\beta)$

Input: two time-distance sequence: δ_α and δ_β
Output: The spatial-temporal difference: ψ

```
Initialization: temp_ψ ← 0; ψ ← 0
while length(δ_α) not equal to length(δ_β) do
    Align and fill between δ_α and δ_β by linear interpolation.
end while
for i = 2; i < length(δ_α); i++ do
    if is Case 1 or Case 2 then
        temp_ψ ← Calculated by equation (8)
    else
        temp_ψ ← Calculated by equation (10)
    end if
    ψ ← ψ + temp_ψ
end for
return ψ
```

An STD between two time-distance sequence δ_α and δ_β can be represented as

$$\psi(\delta_\alpha, \delta_\beta) = \sum_{i=2}^{n} S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}), \tag{7}$$

where $S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)})$ represents the area of the sub-polygon in STD. There are three cases of these sub-polygons, as shown in Fig. 6, trapezoid (case 1), triangle (case 2), and internal triangles of the trapezoid (case 3).

Set the length of AD is p, the length of CB is q, and the distance between AD and CB is h. For the case 1, $S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = S_{ABCD}$, the following equation can be used to calculate the area of trapezoid $ABCD$:

$$S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = \frac{1}{2}(p+q)h. \tag{8}$$

For the case 2, $S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = S_{ABC}$, the area of triangle ABC is

$$S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = \frac{1}{2}qh. \tag{9}$$

It should be noted that, the case 2 is the special case of case 1 when $p = 0$. Therefore, case 2 can also be calculated by Equation (8), where $p = 0$.

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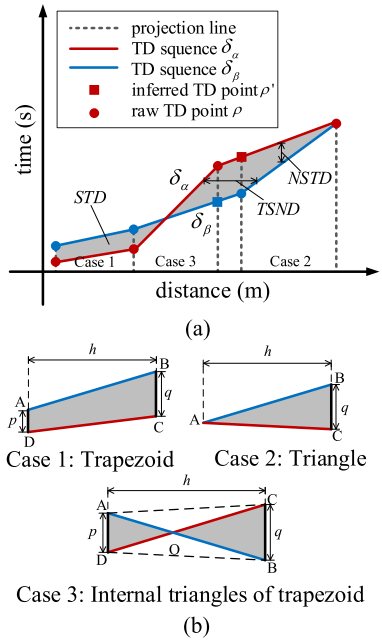

Fig. 6. The illustration of spatial-temporal difference. (a) The illustration of STD. (b) The cases of sub-polygons.

For the case 3, $S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = S_{ADO}+S_{BCO}$, it can be simplified to calculate the sum areas of internal triangle ADO and CBO in the trapezoid.

$$S(\delta_{\alpha(i-1,i)}, \delta_{\beta(i-1,i)}) = \frac{h}{2}\left(\frac{p^2+q^2}{p+q}\right). \tag{10}$$

2) Encode Methods: Encoding compression is a type of method for data character compression, which is orthogonal to OCT and OCT-LSTM methods, and may be combined with each other to further improve the effectiveness of trajectory compression. In this paper, we first convert the floating number into integer (amplify and retain six digit precision), then we utilize Varint, Varnibble, Varbits and Diff encode methods to further compress transmission data of trajectory collection [35], [36].

Varint: The variable length integer (Varint) is a byte-oriented encoding, where each byte holds 7 bit of data and 1 bit is used as the end-of-word marker. Thus smaller values require fewer bytes: 1 byte (<128), 2 bytes (<16384), and so on.

Varnibble: Varnibble encoding is similar to varint, in this case, we operate on 4-bit words instead of 8 bits (each word contains 3 bits of data and 1 bit of end-value marker). This encoding require some bit operations (shift, and, or) but with constant shifts amounts/masks, so this isn't as expansive as other bit-level encoding schemes.

Varbits: Varbits is a generalization of varint and varnibble, which selects the number of bits is done once for the whole sequence, this number is saved on one byte, then all elements are encoded.

Diff: Diff is a method to difference between each value and the first value ("head") are encoded as bit positions, which resolves downside of byte-oriented encoding is wasting bits if values are small.

TABLE I
THE HYPER-PARAMETERS OF OCT-LSTM

hyper-parameters	values
The depth of fully connected layers	3
The depth of LSTM layers	2
The depth of Dense layers	1
The timesteps L	20
The input dimension D_i	1
The first layer LSTM cell dimension D_1	50
The second layer LSTM cell dimension D_2	50
The output dimension D_o	1
The batch size	128
The epochs	10
The validation split	0.2
The dropout	0.2

VI. EXPERIMENT EVALUATION

A. Experiment Setup

1) Dataset: The experiments are based on trajectory data retrieved from two real-life trajectory dataset: GeoLife and PriTra.

GeoLife is a GPS trajectory dataset collected in (Microsoft Research Asia) Geolife project by 182 users in a period of over five years (from April 2007 to August 2012). In order to validate the performance of our methods, we extract the trajectories of 69 users from GeoLife, which have labeled their trajectories with transportation mode as bus, car or taxi. These trajectories are constrained by the road network, with a total length of 53,147 km.

PriTra is a large-scale private car trajectory dataset. By using the location terminal as shown in Fig. 2, we have obtained such a trajectory dataset of more than 40,000 private cars from the Pearl River Delta region in China. In order to validate the performance of our methods, we extract the trajectories of 50 private cars from PriTra, the trajectory periods span six months, the total mileages reach 15,110 km.

2) Hyper-Parameters: The OCT, OCT-LSTM, PRESS and OPERB-A compression methods are implemented and tested via Python code compiled with version 3.5.2. Results are produced on an Intel Xeon E3-1241 v3 processor clocked at 3.5 GHz with 32 gigabytes of RAM, running Windows 10. The activation function of LSTM is tanh, the loss function is mse and the optimizer method is rmsprop. The average training time is around 1052s in the experimental platform, the mean loss value is 1.7800, the mean loss of split validation is 1.2817. Through intensive empirical optimization, we determine the hyper-parameters of the LSTM network as shown in Table I.

3) Metric: In this paper, Compression Ratio (CR) and compression time-delay are used to evaluate the proposed methods. Compression ratio is the key performance metric to measure compression efficiency [28]. Given a raw trajectory T_{raw} and its compressed form T_c, the compression ratio is

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defined as:

$$CR = \frac{|T_{raw}|}{|T_c|}, \tag{11}$$

where $|T_{raw}|$, $|T_c|$ are storage cost of T_{raw}, T_c respectively.

The time-delay of trajectory compression and transmission is mainly caused by three factors: compression time-delay τ_c, transmission time-delay τ_t, and other time-delay τ_o.

$$\tau = \tau_c + \tau_t + \tau_o, \tag{12}$$

where the compression delay τ_c is the delay caused by the compression method, the transmission delay τ_t is the delay caused by the transmission link and the other delay τ_o is the delay caused by other reasons.

4) Baseline: In this paper, we choose two methods PRESS (paralleled road-network-based trajectory compression) and OPERB-A (aggressive one-pass error bounded) as the baseline for comparison.

PRESS [25] is a map-matching based trajectory compression method, which represents a given trajectory using a spatial path and a temporal sequence and then employs Hybrid Spatial Compression (HSC) algorithm and error Bounded Temporal Compression (BTC) algorithms to compress the spatial path and temporal sequence, respectively. This is a classic trajectory compression method based on map matching, so it is selected as a baseline for comparison.

OPERB-A [37] is an aggressive one-pass error bounded trajectory simplification algorithm (OPERB-A), which scans each data point in a trajectory once and only once, and allows interpolating new data points into a trajectory under certain conditions. This is a novel online compression method that sufficient the needs of sensors in mobile devices, so it is chosen as a baseline for comparison.

The performance evaluation selects the same trajectory data input. After the same pre-processing, STD is used as the error metric, and then the experiment is compared.

B. Compression Ratio of Spatial-Temporal Transformation

In the first set of experiments, we discuss the effectiveness of spatial-temporal transformation stage. In the spatial-temporal transformation stage (Section V-C), T_{raw} is divided into TD sequence δ and path sequence $Path$, and some trajectories uncertainty is eliminated through discarding the Stop Points (SPs) and shape filling by the road network. Fig. 7 shows the compression ratio (CR) after SPs discarding, shape filling and the entire preprocessing, where the horizontal coordinate is the ID of the private cars, and the ordinate is the compression ratio. In order to make the display of Fig. 7 more intuitive, the private ID values are sorted according to the CR of the entire preprocessing. The blue bars are the compression ratio after SPs discarding. Each blue bar represents the average compression ratio of a private car after SPs discarding. These differences are determined by driving habits and traffic information. As you can see from the bars of SPs discarding, the maximum CR is 2.73, the minimum CR is 1.04 and the mean CR is 1.48, this indicates 32.43% of $T_{matched}$ are stay points, which is redundant and should be discarded. Similarly, the orange bars are the compression ratio curve after shape filling. Each orange bar represents the average compression ratio of a private car after the shape filling. The shape filling step added some missing information by road network data. As you can see from the CR curve of shape filling, the maximum CR of the shape filling is 0.96, the minimum CR is 0.93 and the mean CR is 0.94. These value of CR are smaller than 1, its means that the missing road network information has been inserted about 4% to 7% after shape filling. Moreover, the amount of information added by shape filling depend on road network details. In practice, we can choose whether to proceed with the shape filling based on the service requirements. The black curve with dots is compression ratio curve of the entire preprocessing (or spatial-temporal transformation stage), as shown in Fig. 7, where the mean CR of entire preprocessing is 1.4, the maximum CR is 2.58, and the minimum CR is 0.98. This is consistent with the experimental results of Stop Points (SPs) and shape filling.

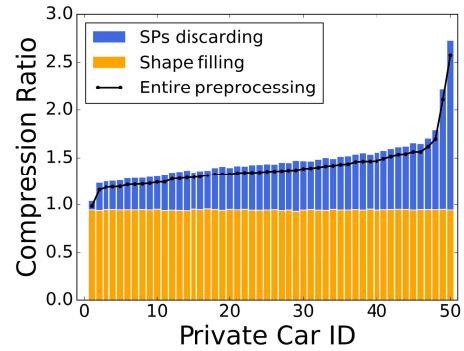

Fig. 7. The CR of SPs discarding and Shape filling.

C. Experimental Comparison of Compression Ratio

In the second set of experiments, we first compared the average compression ratio of OCT-LSTM under different maximum allowable STD error ψ. As shown in Fig. 8, the horizontal coordinate is maximum allowable STD error ψ, and the ordinate is compression ratio, where the red horizontal line represents the mean value of CR, and red square point represents the median of CR. When the maximum STD error ψ is $(0.01, 0.25, 1, 4, 25)$, the mean of CR is $(4.76, 16.05, 28.49, 51.97, 127.3)$, the median of CR is $(4.80, 15.74, 28.19, 51.22, 123.25)$ and the variance of CR is $(0.96, 3.46, 5.99, 10.65, 24.27)$. The experimental results show that when STD error ψ increases, the compression ratio increases rapidly.

Then we compare four methods: PRESS, OPERB-A, OCT and OCT-LSTM. PRESS is a map-matching based trajectory compression method [28]. OPERB-A is an online aggressive one-pass error bounded trajectory simplification algorithm which sufficient the needs of sensors in mobile devices [37]. OCT and OCT-LSTM are the methods proposed in this paper, OCT is an opportunity compression method based on the mean fitted trajectory, OCT-LSTM is an opportunity compression method based on LSTM. As shown in Fig. 9, the horizontal coordinate is the maximum allowable STD error ψ, and the ordinate is compression ratio. Fig. 9(a) shows the comparison of compression ratio in PriTra dataset. Fig. 9(b) shows the

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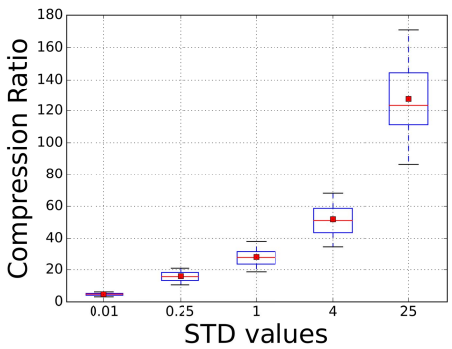

Fig. 8. The CR of OCT-LSTM in different STD ψ.

comparison of compression ratio in GeoLife dataset. Experimental comparisons in these two datasets show that OCT-LSTM gets the best compression ratio, even if the OCT method is better than the OPERB-A, and PRESS method. OCT and OCT-LSTM are able to achieve much better compression ratios than PRESS when ψ from 4 to 100. When ψ in the range of $[0.01, 4]$ the OCT, OPERB-A and PRESS have similar compression ratio. However, when the ψ error is set to 100, OCT and OCT-LSTM can get much larger of compression ratio than OPERB-A and PRESS, which is considered to reduce transmission cost and data size on trajectory collection. This is because the OCT and OCT-LSTM are to adopt a prediction-based strategy to reduce the synchronization times of trajectory data. When the predicted value of the model is close to the true value, the data center can directly use the predicted value instead of the transmission of the true value, thus the transmission cost is reduced. This is the main reason for the higher compression ratios of OCT and OCT-LSTM. The experimental results of CR show that the compression strategy based on the prediction model of OCT and OCT-LSTM can achieve higher trajectory compression ratio than the traditional compression methods (such as PRESS and OPERB-A). This strategy of improving compression ratio by prediction model is novel and different from traditional compression methods. By using the LSTM model to predict the TD sequence of the trajectory, the OCT-LSTM has a higher prediction accuracy than the OCT prediction model, so it can obtain a higher CR.

Finally, we separately compare the compression effect of encoding compression on the data. These encoding methods are orthogonal to OCT and OCT-LSTM methods, and we can choose whether to combine encoding compression according to the application requirements. As shown in Table II, 32-bit int is the method convert 32-bit floating numbers to 32-bit integers, the CR is 1 means that the data for this step is not compressed. The CR of Varint, VarBits, Varnibble is $(1.82, 1.84, 1.75)$, and the CR of Varint, VarBits, Varnibble combine Diff is $(3.85, 3.87, 3.81)$, these encoding compression methods can compressed data furtherly, and their runtime is very small (less than 0.01ms) can be ignored.

D. Experimental Comparison of Time-Delay

In the third set of experiments, we first statistics the transmission time-delay τ_t and other time-delay τ_o in PriTra dataset,

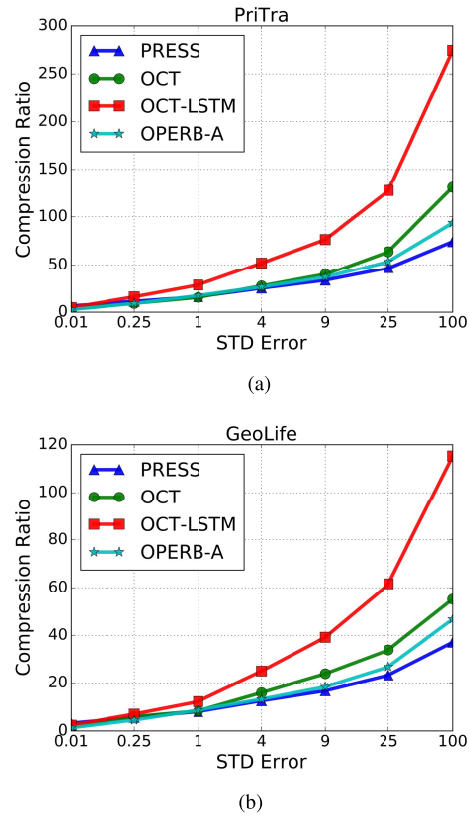

Fig. 9. The compression ratio of different methods in PriTra/GeoLife dataset. (a) Compression ratio comparison in PriTra dataset. (b) Compression ratio comparison in GeoLife dataset.

TABLE II
COMPRESSION RATIO COMPARISON OF ENCODING METHODS

Encode methods	Compression ratio
32-bit int	1
Varint	1.82
Varint [diff]	3.85
VarBits	1.84
VarBits [diff]	3.87
Varnibble	1.75
Varnibble [diff]	3.81

which reflects the time-delay in real-time trajectory data collection when trajectory compression is not performed. This statistic cannot be performed in GeoLife, because GeoLife data does not record the received timestamp of trajetory. Set t_r is the timestamp of data received by the server, t_c is the timestamp of data collected by the GPS module, then the $\tau_t + \tau_o = t_r - t_c$. Due to the lack of t_r data in Geolife dataset, only statistics of the time-delay $\tau_t + \tau_o$ in PriTra dataset are given in this paper, as shown in Fig. 10, about 81.77% time-delay is in $0 \sim 1s$, about 9.96% time-delay is in $1 \sim 10s$, about 6.15% time-delay is in $10 \sim 60s$, and about 2.12% time-delay is more than $60s$.

Then we compare the average compression time-delay τ_c of PRESS, OPERB-A, OCT, and OCT-LSTM in both PriTra and GeoLife dataset. As shown in Fig. 11, the horizontal

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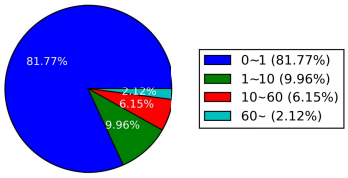

Fig. 10. The statistics pie chart of time-delay $\tau_t + \tau_o$ data in PriTra.

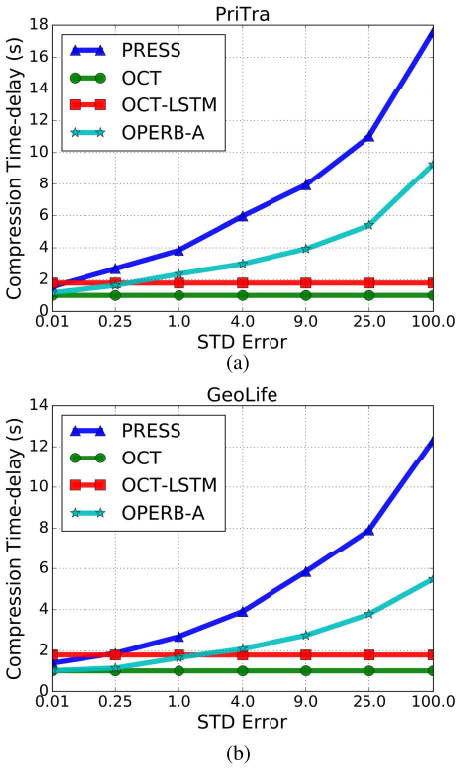

Fig. 11. The compression time delay of different methods in PriTra/GeoLife dataset. (a) Compression time delay comparison in PriTra dataset. (b) Compression time delay comparison in GeoLife dataset.

coordinate is the maximum allowable STD error ψ, and the ordinate is compression time-delay. Fig. 9(a) shows results in PriTra, Fig. 9(b) shows results in GeoLife, respectively. It is obvious that our approach is able to achieve a much lower time-delay than OPERB-A and PRESS. The mean time-delay in OCT method is stable around 1 second (limited by sampling interval of raw trajectory). Similarly, mean time-delay in OCT-LSTM method is stable around the data upload interval 1.81 second (limited by sampling interval of raw trajectory and computing time of LSTM Model). However, the compression time-delay of PRESS and OPERB-A get larger when STD error ψ grown (limited by the information update mechanism of trajectory simplify method). In the OCT and OCT-LSTM strategy, each sampling interval will compare the predicted value with the real value, and judge whether the prediction error needs to be updated or not. This mechanism can update the trajectory information in near real time. The traditional simplified trajectory compression (such as PRESS, OPERB-A) does not complete the compression until the next update, resulting in a high compression delay. This is the main reason why the compression delay τ_c of OCT and OCT-LSTM is lower. As a summary, our approach is very effective in terms of real-time compressing, which is the main objective of the compression framework proposed in this work. This strategy of OCT and OCT-LSTM, which is based on two synchronous prediction models to reduce trajectory transmission, not only ensures the real-time transmission of trajectory information but also achieves better compression efficiency, which is novel and different from the traditional trajectory compression method. The experimental results will demonstrate the power of our framework and hence justify the design.

VII. Conclusion

In this paper, to reduce the transmission overhead of private car trajectory data, we propose OCT-LSTM, an Opportunistic Compression and Transmission approach via Long Short-Term Memory networks. OCT-LSTM employs novel opportunistic compression strategy to compress the spatial and temporal information of the trajectories. We test the performance of OCT-LSTM based on real-life private cars trajectory dataset. Results reveal that the proposed OCT-LSTM outperforms the state-of-the-art compression framework in terms of compression ratio and compression delay-time. Moreover, OCT-LSTM is able to achieve real-time compression for trajectory data flows. Although the trajectory compression based on OCT-LSTM is run on a desktop in our experiments, we plan to implement this framework on real vehicle location terminals by integrating a small size ROM for storing digital map and a low-cost microprocessor in the future.

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Jie Chen received the B.S. degree from the Hunan University of Science and Technology, China, in 2010, the M.S. degree in computer science and technology from Hunan University, China, in 2015, where he is currently pursuing the Ph.D. degree in computer science and technology. His research interests include interests include data mining, artificial intelligence, and intelligent transportation system.

Zhu Xiao (M'15) received the M.S. and Ph.D. degrees in communication and information systems from Xidian University, China, in 2007 and 2010, respectively. From 2010 to 2012, he was a Research Fellow with the Department of Computer Science and Technology, University of Bedfordshire, U.K. He is currently an Associate Professor with the College of Computer Science and Electronic Engineering, Hunan University, China. His research interests include wireless localization, mobile computing, and heterogeneous networks.

Dong Wang (M'15) received the B.S. and Ph.D. degrees in computer science from Hunan University, China, in 1986 and 2006, respectively. From 2004 to 2005, he was a Visiting Scholar with the University of Technology Sydney, Sydney, Australia. Since 1986, he has been with Hunan University, where he is currently a Professor. His main research interests include network test and performance evaluation, wireless communications, and mobile computing.

Daiwu Chen received the B.S. degree in computer applications from the National University of Defense Technology, China, in 1997, and the M.S. degree in software engineering from Wuhan University, China, in 2005. Since 1986, he has been with the Hunan University of Humanities, Science and Technology, China, where he is currently an Associate Professor. His research interests include computer network, multi-media technology, and mobile computing.

Vincent Havyarimana received the B.S. degree in mathematics from the University of Burundi, Bujumbura, in 2007, and the M.E. and Ph.D. degrees in computer science and technology from Hunan University, Changsha, China, in 2011 and 2016, respectively. He is currently a Lecturer and the Head of Research Service with the Ecole Normale Supérieure Burundi. His research interests include wireless communication, vehicular ad hoc networks, and mobile computing.

Jing Bai (M'10) received the B.S. degree in electronic and information engineering from Zhengzhou University and the Ph.D. degree in pattern recognition and intelligent system from Xidian University in 2004 and 2009, respectively. She is currently an Associate Professor with Xidian University. Her research interests include image processing, machine learning, and intelligent information processing.

Hongyang Chen (M'11–SM'16) received the Ph.D. degree from The University of Tokyo. In 2009, he was a Visiting Researcher with the UCLA Adaptive Systems Laboratory, University of California at Los Angeles. His research interests include IoT, intelligent networking, location estimation and location-based service, and signal processing for communications. He served as an Editor for the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS and as the Associate Editor for the IEEE COMMUNICATIONS LETTERS.

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